Enhanced Automated Construction Quantity Takeoff: Integrating LLMs with Data Extraction, Symbol Mapping, and Rule-based Reasoning

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Abstract

Traditional quantity takeoff (QTO) methods, which rely on manual processes and 2D drawings, are time-consuming, labor-intensive, and prone to human errors. Though building information modeling (BIM) has introduced improvements using digital models, its application is often limited to projects where BIM models are readily available. In scenarios where traditional construction documents are the primary source of information, there remains a significant challenge in automating QTO processes. To address the issue, this study proposes a novel approach that leverages large language models (LLMs) to automate QTO directly from these traditional 2D documents. Our approach enhances general LLMs by integrating advanced data extraction, precise symbol mapping and structured chain-of-thought (CoT) reasoning according to domain-specific rules. This method significantly improves the accuracy and efficiency of QTO tasks by enhancing the ability of LLMs to understand human language, offering a robust solution of QTO tasks in contexts where BIM is not feasible. We validated our approach through a case study on excavation tasks. The results showcased the shortcomings of applying general LLMs like GPT-4o solely, which might raise significant errors in QTO tasks. On the contrast, the proposed method demonstrated precise and reliable calculation with a complete chain of explainable reasons.

Keywords: Quantity takeoff; Large language models; Automated data extraction; Symbol mapping; Chain of thought; Rule-based reasoning

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1. Introduction

Quantity takeoff (QTO) is a fundamental task in construction project management [1], forming the basis for cost estimation, procurement, and scheduling [2]. Traditionally, QTO has been a manual process relying heavily on 2D drawings and manual calculations [5]. Although widely used, this method is time-consuming, prone to human errors [6], and often leads to inaccuracies, especially in large and complex projects. In recent years, building information modeling (BIM) has been introduced to enhance the QTO process [7]. BIM enables the creation of data-rich, intelligent models that link individual building elements to their respective materials properties [8], significantly improving the speed, accuracy, and reliability of QTO.

Despite the advancements brought by BIM, many construction projects continue to rely on traditional construction documents [9]. This reliance is due to the social and habitual resistance to change of implementing BIM [10], the unavailability of comprehensive BIM models for some projects, and established documentation practices.

Recent advances in natural language processing (NLP) and the development of large language models (LLMs) offer new possibilities for automating text-based tasks, including document review, compliance assistance, and report generation in the construction industry [12]. In practice, however, automating QTO from traditional construction documents presents unique challenges. Traditional documents, although comprehensive, present significant challenges for automated QTO due to their unstructured nature and variability [13]. Automating QTO from these documents requires handling inconsistent formats and diverse terminology, thus complicating the extraction and accurate calculation of quantities [14].

To fulfil this gap, this research proposes a novel methodology that leverages LLMs to automate QTO calculations directly from construction documents. By focusing on extracting relevant data, mapping corresponding symbols, and employing advanced

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rule-based reasoning methods, this study aims to develop an automated pipeline that enhances both the efficiency and reliability of QTO processes.

The structure of this paper is as follows: Chapter 2 reviews existing research on traditional QTO methods and the applications of LLMs in the construction industry, highlighting the remaining challenges and research gaps. Chapter 3 presents the proposed methodology, detailing the processes of rule parsing, data extraction, symbol mapping and advanced reasoning methods. Chapter 4 showcases a case study to validate the effectiveness of the proposed methodology. Finally, Chapter 5 discusses the limitations of proposed method and potential future research directions, and concludes the paper by summarizing the key contributions.

2. Related work

This chapter reviews existing research on traditional QTO methods and the applications of LLMs in the construction industry. It also discusses the remaining challenges and research gaps, emphasizing the need for further advancements and innovations in this area.

2.1. Traditional quantity takeoff

In early construction projects, QTO processes were manual, relying on 2D drawings and manual calculations. While common, this approach is time-consuming, error-prone, and can lead to inaccuracies, particularly in complex projects. BIM has transformed QTO by enabling the creation of detailed, intelligent models that associate specific building components with their materials properties, significantly enhancing the efficiency, precision, and dependability of the QTO process.

Several studies have highlighted the advantages of using BIM for QTO. Hartmann et al. [15] discussed the alignment of BIM tools with construction management methods, highlighting improved efficiency in data management and decision-making. Choi et al.

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[16] introduced an Open BIM-based QTO system for early design stages, improving the reliability of estimations. Plebankiewicz et al. [17] developed a BIM-based system for quantity and cost estimation, showcasing how BIM can facilitate more accurate and detailed cost assessments. Zima [18] investigated the impact of BIM data on QTO results, stressing the importance of accurate modeling for reliable quantity estimates. Similarly, Kula et al. [19] demonstrated the potential of BIM for automated QTO, emphasizing its ability to streamline the QTO process.

However, despite these demonstrated benefits, several studies have also highlighted the limitations and challenges associated with BIM-based QTO. Monteiro and Martins [20] noted that the insufficient adaptability of BIM tools to varied design settings could limit user expectations and result in inadequate quantity estimations. Kim et al. [21] analyzed discrepancies in quantities extracted from different BIM components, offering strategies to minimize these inconsistencies. Khosakitchalert et al. [22] focused on enhancing the accuracy of BIM-based QTO for compound elements, such as walls and floors, by emphasizing the need for detailed and precise modeling methods. These studies collectively underscore that while BIM-based QTO significantly improves the efficiency and accuracy of the QTO process, it also presents challenges related to model quality, data integration, and consistency checks.

2.2. Large language models in construction

With advances in NLP technology, LLMs are demonstrating strong capabilities across various industries [23]. Recently, the construction industry has increasingly adopted LLMs to address multiple challenges and enhance efficiency across different domains. The applications of LLMs in construction can be broadly categorized into text-based tasks and multimodal tasks.

In text-based tasks, LLMs have shown significant potential in automating and improving various aspects. Kim et al. [24] developed an AI assistant that helps engineers navigate complex civil engineering codes and standards, improving

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compliance and reducing the time required for manual interpretation. Similarly, Cruz-Castro et al. [25] introduced an LLM-based system designed to provide real-time review and feedback for technical documents in urban construction and technology. This system enhances the quality and consistency of technical writing, aiding professionals in maintaining high standards in their documentation. Domain-specific LLMs trained on construction management materials have also shown promise in improving the relevance and accuracy of language model outputs. Zhong and Goodfellow [26] highlighted the effectiveness of these models in automating construction management tasks, emphasizing the need for domain-specific training to maximize the utility of LLMs.

Multimodal applications of LLMs in construction involve the integration of language models with other modalities to tackle complex problems. Pu et al. [27] developed the AutoRepo framework, which uses multimodal LLMs to automate the creation of construction reports. This framework integrates diverse data sources and employs advanced language processing to generate comprehensive reports, significantly reducing manual effort and improving accuracy. Sun et al. [28] investigated the use of vision and language models to recognize construction waste materials. This approach aids in the efficient sorting and recycling of construction waste, contributing to more sustainable construction practices. Yao and de Soto [29] utilized language models to enhance cybersecurity within the construction industry by identifying cyber risks, providing a proactive approach to mitigating potential threats and ensuring the security of construction projects. Jung et al. [30] demonstrated the integration of LLMs with vision models to map construction schedule activities to Uniformat categories, facilitating better project management and resource allocation.

2.3. Research gap

Although BIM has made significant research progress in improving QTO processes, research on applying LLMs to QTO automation is still very scarce in the

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existing literature. Existing research mainly focuses on improving the efficiency and accuracy of QTO through BIM, while no research explores how to utilize LLM to automate the QTO process of traditional construction documents. Therefore, this study fills this academic gap and proposes an automated QTO scheme that combines LLMs with domain-specific rules, OCR technology, and chain reasoning, providing a new alternative to existing BIM-dependent methods.

3. Proposed method

This chapter describes in detail our proposed method for automating QTO calculation. The method is based on parsing QTO rules, data extraction and symbol mapping, and the chain-of-thought (CoT) reasoning method. The aim is to enhance the efficiency and accuracy of the QTO calculation process. The following sections will introduce each key step in detail, explaining the specific implementation and technical details.

3.1. Parsing quantity takeoff rules

In this section, the process of parsing QTO rules and converting them into standardized formula calculation chains is described. This step ensures the accuracy of symbol mapping in subsequent steps by systematizing and standardizing QTO rules, laying the foundation for the automated QTO process.

3.1.1. Introduction of external quantity takeoff rules

In construction projects, the takeoff process involves detailed calculations to determine the quantities of materials and labor required [31]. However, the current lack of LLMs specifically tailored for the civil engineering domain results in a significant gap in domain-specific knowledge. To address this issue, it is essential to incorporate external knowledge bases, such as rule documents for QTO calculations. These

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documents provide standardized guidelines and formulas for various construction tasks, which can be used to systematize and automate the QTO process.

The external documents for QTO calculations typically include both textual descriptions and mathematical formulas. Firstly, the rule documents are input into multimodal LLMs. The textual descriptions are then processed by the LLM to understand their semantic meaning. When explicit formulas are provided, the LLM recognizes and interprets these formulas. In cases where explicit formulas are not provided, the LLM extracts the corresponding formulas from the textual descriptions. Finally, the parsed rules and formulas are integrated into LLMs, enhancing its ability to perform accurate and efficient QTO calculations. This process is illustrated in Figure 1.

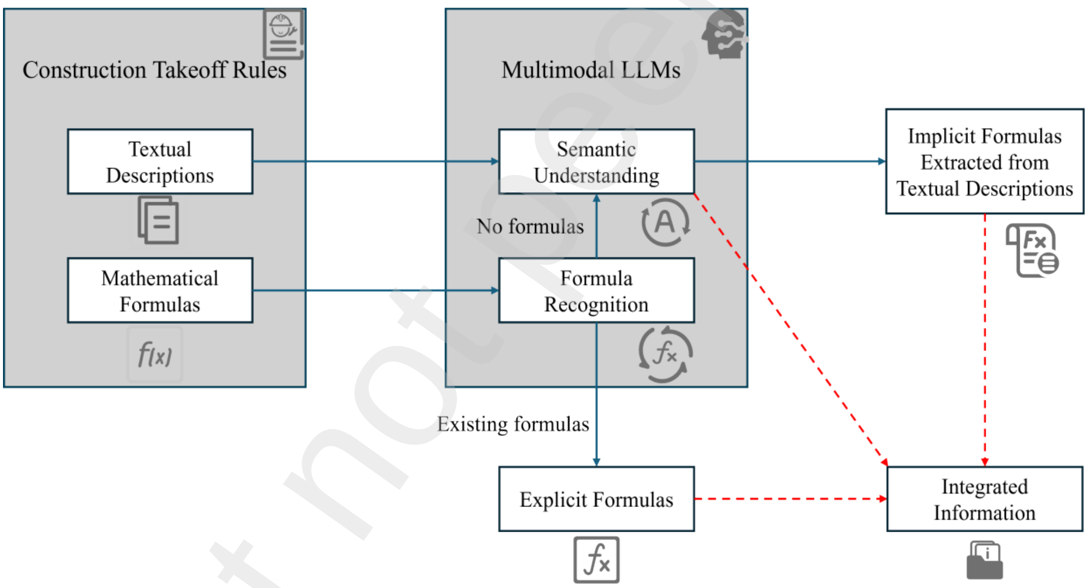

Figure 1: Pipeline for parsing QTO rules using multimodal LLMs

3.1.2. Construction of formula calculation chains

Once the QTO rules have been parsed and variable symbols have been assigned to the textual elements, the next critical step is to create standardized formula calculation chains. In principle, rules are hierarchical and follow a chain structure, which is consistent with the way human logic proceeds sequentially. By organizing the parsed and standardized formulas into a chain, this natural logical flow is reflected, where each

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step leads directly to the next in a clear, orderly sequence. This chain structure not only enhances the clarity and consistency of the calculations but also ensures that the interdependencies and progression of the QTO tasks are accurately represented. Unlike more complex networks, which can obscure the flow of information, or single scalar representations, which might oversimplify the process, a chain offers the most intuitive and effective way to model the logical steps required for precise QTO calculations.

The QTO rules, now represented with variable symbols and standardized formulas, need to be linked together to form a coherent calculation chain. This involves ensuring that the output of one formula can be used as input to another formula, depending on the order of QTO calculations and the corresponding variable symbols. This linkage is crucial for maintaining the accuracy and consistency of the QTO calculations throughout the project lifecycle.

A typical example is the estimation of material quantities for earthwork, which can be calculated through a three-step chain:

1. **Calculation of Compaction Factor:**

$$CF = LT \ / \ CT \tag{1}$$

Where LT is Loose Thickness and CT is Compacted Thickness.

2. **Calculation of Adjusted Measurement Unit**:

$$AMU = MU \times CF \tag{2}$$

Where MU is Measurement Unit and CF is Compaction Factor from the first step.

3. **Calculation of Estimated Quantity**:

$$EQ = AMU \times (1 + LR) \tag{3}$$

Where AMU is Adjusted Measurement Unit from the second step and LR is Loss Rate.

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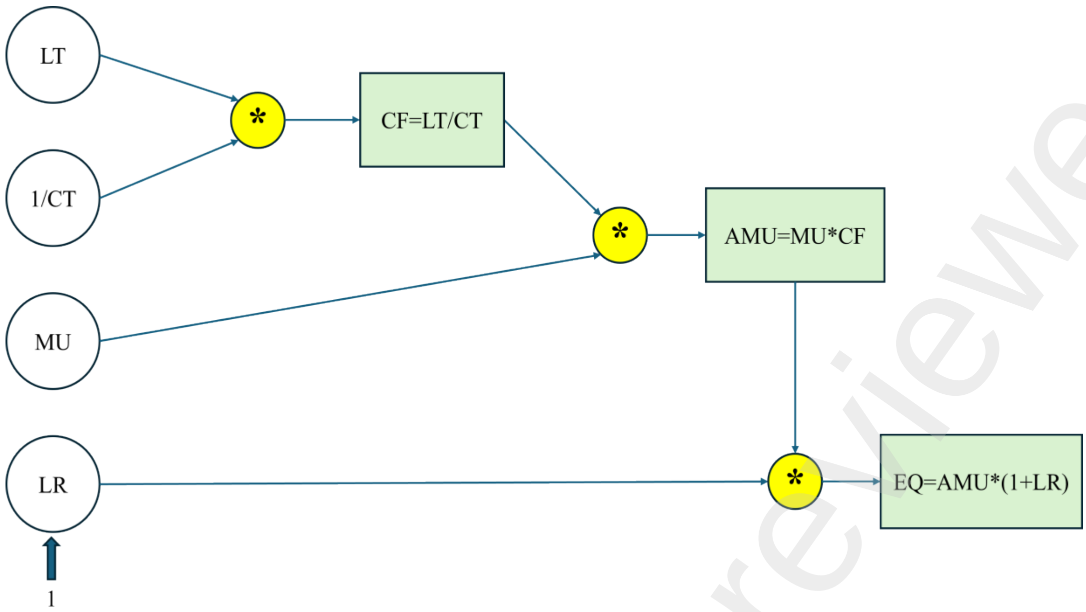

Figure 2: Example of QTO calculation chain

In this example, the calculation chain starts with determining the compaction factor using loose and compacted thickness. The next step involves adjusting the measurement unit using the compaction factor. Finally, the estimated quantity is computed using the adjusted measurement unit and the loss rate. This structured approach allows for automated calculation of material quantities based on the parsed rules and extracted data. This process is illustrated in Figure 2.

By systematically converting parsed rules into interconnected sequences of formulas, a robust calculation chain for automating the QTO process is established. This approach minimizes errors, enhances efficiency, and ensures that QTO calculations are both accurate and consistent, thereby improving project planning and resource management.

3.2. Multi-source data extraction and recognition

In the previous section, the calculation chain required to perform exact QTO calculation was established. However, the variability in requirements and parameters across different construction projects [32] introduces significant challenges. These challenges arise because each project may involve unique data formats, measurement

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units, and contextual conditions that can complicate the automated QTO process. To address this, our method involves extracting and mapping specific symbol values from project-specific data sources, ensuring that the automated QTO calculations are not only accurate but also tailored to the distinct needs of each project. This approach is crucial for enabling our method to adapt to diverse project contexts, thereby maintaining the precision and reliability of QTO calculations across various scenarios.

In this section, the process of extracting data from various document sources is explained in detail. Accurate data extraction is critical for ensuring that the information contained within different types of construction documents such as PDFs, Word files, and Excel spreadsheets can be effectively utilized in the automated QTO calculations. To achieve this, advanced Optical Character Recognition (OCR) technologies [33] are employed. These technologies are designed to identify and extract data from diverse document formats with high precision. The process begins with segmenting input documents into distinct components [34], such as tables, text, and images, which are then processed by different OCR technologies to extract multiple sets of machine-readable construction data.

However, OCR outputs, particularly from documents with complex layouts or poor quality, often contain inaccuracies that can impact subsequent data processing steps. To address these challenges, we implement a self-correction mechanism based on LLMs. After the initial OCR stage, the extracted data goes through a self-correction process where LLM analyzes the identified text and numerical data to identify and correct potential errors. This self-correction involves cross-referencing extracted portions of the data with known original documents to ensure consistency and accuracy. This approach leverages the LLM's contextual understanding and reasoning capabilities to refine the data, improving the reliability of the information extracted from diverse document formats.

As depicted in the Figure 3, from layout analysis and OCR processing to LLM-based self-correction, this multi-step process ensures the final output contains accurate

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construction data, which is critical for accurate automated QTO and other downstream construction management tasks.

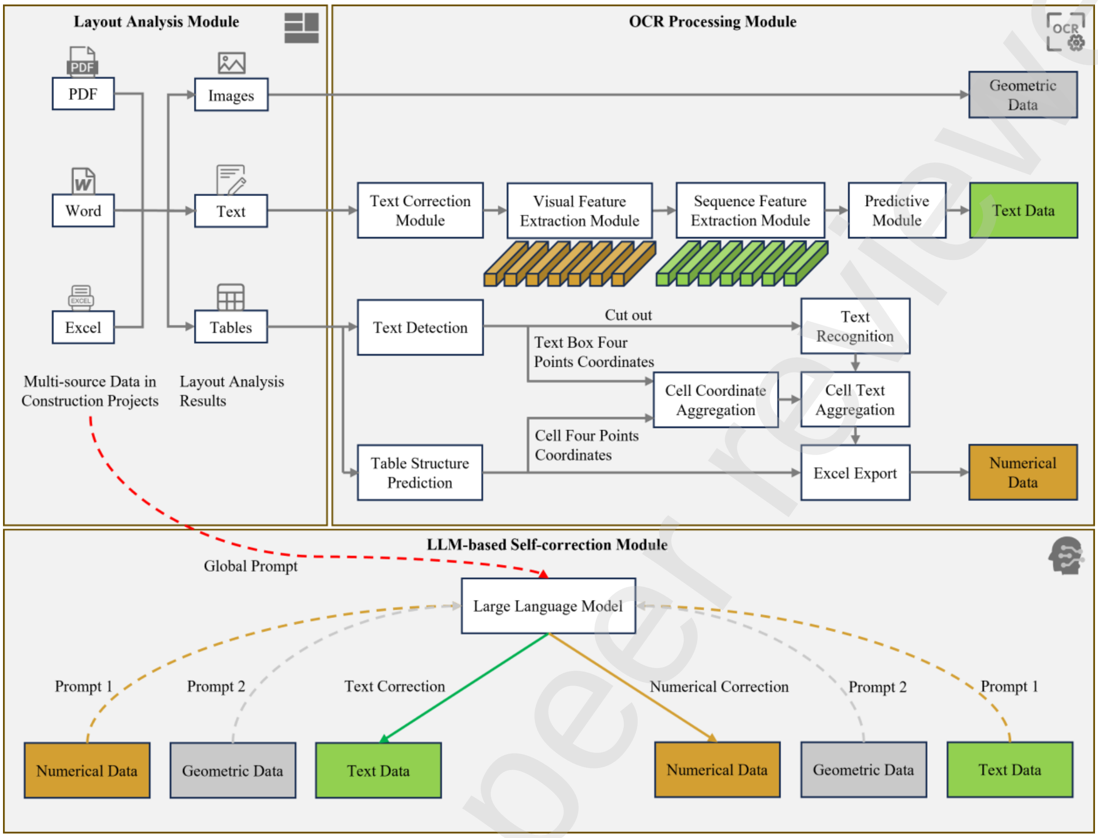

Figure 3: Multi-source data extraction and recognition process

3.3. Symbol mapping and integration

Once the multi-source data has been extracted from various sources, the next step is to map these data to the corresponding symbols defined in the QTO rules. Symbol mapping ensures that each piece of extracted data is correctly associated with its respective variable in the formula calculation chains.

Given the powerful data analysis capabilities, LLMs are used to process the numerical data, text data, and geometric data extracted from the multi-source documents. LLMs leverage rich information from the construction data, integrated information from previously parsed external specifications, and established calculation chains to match data with symbols in the calculation chain. This involves using the text data, geometric data and integrated information as references to accurately associate

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numerical data with the corresponding symbols in calculation chains. By combining all relevant information, LLMs ensure precise and accurate mapping of all extracted data to their respective symbols.

This integrated approach is illustrated in Figure 4, where multi-source data is processed and mapped to the QTO calculation chains using LLMs. By ensuring that each piece of data is accurately captured and correctly associated, the accuracy and efficiency of the QTO process can be significantly improved.

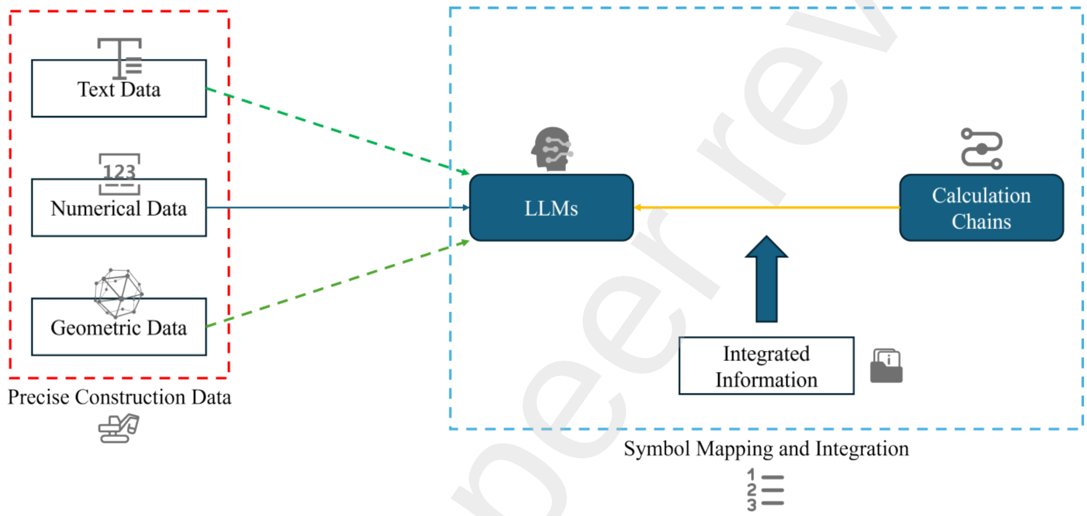

Figure 4: Symbol mapping and integration process

3.4. Chain of Thought reasoning

In this section, we introduce the CoT reasoning method and its application in the context of automated QTO calculations. The CoT reasoning method leverages the capacity of LLMs to perform complex reasoning tasks by generating a sequence of intermediate reasoning steps that lead to the final answer [35]. This method enhances the accuracy and efficiency of the automated QTO process by systematically breaking down complex calculations into manageable steps.

3.4.1. Introduction of CoT reasoning

CoT reasoning involves breaking down complex calculations into a sequence of intermediate steps that guide the model through a logical progression [36]. This method

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leverages the capabilities of LLMs to perform detailed, step-by-step reasoning, enhancing the accuracy and efficiency of complex tasks.

The CoT reasoning process can be formalized as follows:

Given a complex problem P that needs to be solved, CoT reasoning breaks P into a sequence of sub-problems $S_1, S_2, \cdots, S_n$, where solving each sub-problem S_i leads to the final solution. Mathematically, if f represents the function to solve P, then:

$$f(P) = f_n(\cdots f_2(f_1(P))) \tag{4}$$

Each function f_i represents an intermediate step, transforming the problem incrementally until the final solution is achieved.

3.4.2. Automated prompt template generation based on QTO rules

The initial step in implementing the CoT reasoning method involves the generation of automated prompt templates based on QTO rules. This process is crucial as it provides the foundation for accurate and efficient reasoning in subsequent steps.

First, the parsed and standardized QTO rules are integrated into a structured calculation chain. This calculation chain provides a logical sequence of steps required to perform accurate QTO calculations. By systematically linking parsing rules, it is possible to ensure that the output of one calculation is used as input for the next, thus maintaining consistency and accuracy throughout the process.

Next, this calculation chain is applied to project-specific data. This involves extracting the necessary data from multi-source project documents and mapping them to the symbols defined in the calculation chain. For example, using project-specific data such as:

- Loose Thickness (LT) = 1.2 meters
- Compacted Thickness (CT) = 1.0 meter

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- Measurement Unit (MU) = 100 cubic meters
- Loss Rate (LR) = 0.05

The structured calculation chain and project data are then used to generate the prompt template. This template is formulated as shown in the Figure 5.

By following this structured prompt template, the model is guided through each step of the calculation chain. This approach ensures that the model can accurately follow the sequence of calculations, demonstrating the practical application and utility of constructing these chains.

In summary, the automated prompt template generation based on QTO rules, supported by structured calculation chains, accurate data extraction and symbol mapping, ensures that LLMs are equipped with the necessary instructions to perform precise and efficient QTO calculations.

Using the established calculation chain, apply the project-specific data to compute the required quantities.

Project Data:
- Loose Thickness (LT): 1.2 meters
- Compacted Thickness (CT): 1.0 meter
- Measurement Unit (MU): 100 cubic meters
- Loss Rate (LR): 0.05

Calculation Chain:
1. Calculate the Compaction Factor (CF):
   CF = LT / CT
2. Calculate the Adjusted Measurement Unit (AMU):
   AMU = MU * CF
3. Calculate the Estimated Quantity (EQ):
   EQ = AMU * (1 + LR)

Steps:
Step 1: What is the Compaction Factor (CF) if LT is 1.2 meters and CT is 1.0 meter?
CF = LT / CT
Step 2: What is the Adjusted Measurement Unit (AMU) if MU is 100 cubic meters and CF from step 1?
AMU = MU * CF
Step 3: What is the Estimated Quantity (EQ) if AMU from step 2 and LR is 0.05?
EQ = AMU * (1 + LR)

Calculate each step and provide the final estimated quantity of material required.

Figure 5: Example of prompt template for QTO calculation

3.4.3. Dynamic prompt generation and calculation

Dynamic prompt generation refines and adapts the pre-established prompt

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templates to accommodate project-specific data and contextual variables, ensuring the accuracy and relevance of QTO calculations performed by LLMs.

The process begins with the integration of project-specific data into the pre-established prompt templates. This data, which includes measurements, material properties, and other relevant variables, is extracted and standardized to match the template structure. By dynamically filling the placeholders in the templates, prompts are created precisely customized for each project.

The dynamically generated prompts guide LLMs through a sequence of calculations, where each step uses the integrated project-specific data to compute intermediate results for subsequent steps. This structured approach ensures that LLMs perform calculations accurately and efficiently, handling the variability in project data. Detailed reasoning steps are included in the prompts to break down the overall task into manageable sub-tasks, providing a clear and logical progression from one calculation to the next. This process is shown in Figure 6.

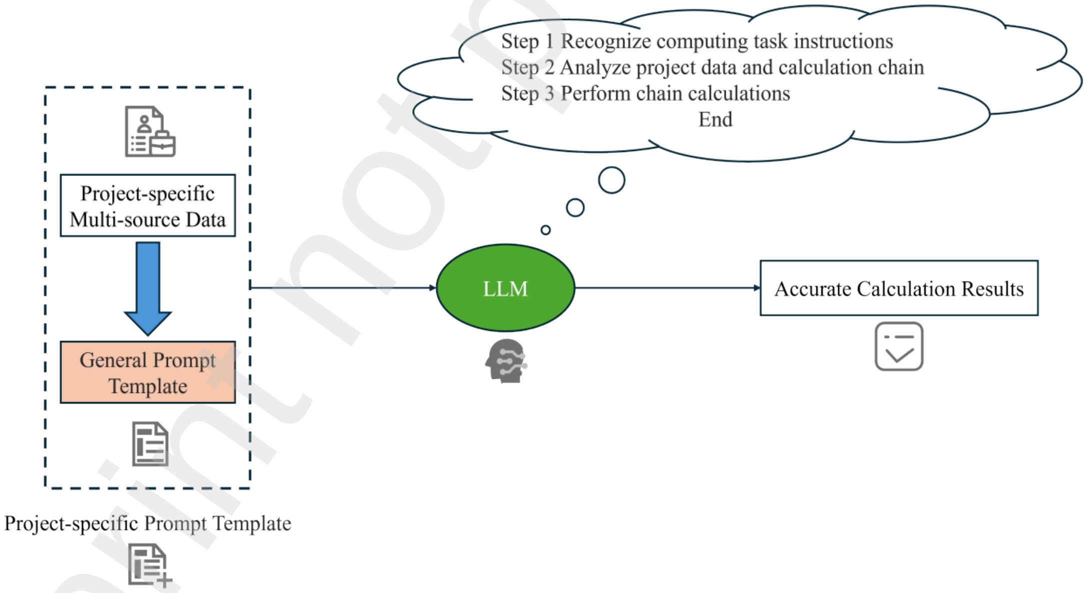

Figure 6: Dynamic prompt generation and calculation workflow

In summary, dynamic prompt generation and calculation enhance the capability of LLMs by ensuring that prompts are contextually relevant and accurately reflect the specific conditions of each project. This approach enables the models to perform precise and efficient QTO calculations, tailored to the unique requirements of each project,

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resulting in optimized resource allocation and improved project outcomes. Additionally, the reuse of general prompt templates effectively reduces the inference load on LLMs, making the process more efficient and resource-saving.

4. Case study

To validate the proposed method, we have conducted a case study that demonstrates the application and effectiveness of automating QTO calculations directly from construction documents using LLMs, specifically GPT-4o [37], currently the most advanced LLM in the world. This case study focuses on the common QTO task of excavation [38].

We begin by introducing an example problem related to excavation and initially use GPT-4o to perform the QTO calculations without any preprocessing or specialized methodologies. Next, we apply our proposed method to the same problem. Our approach involves parsing the relevant construction specifications, generating the appropriate formulas and calculation chains, extracting the necessary data using advanced OCR technologies, and performing symbol mapping and integration. By employing the CoT reasoning method, we guide GPT-4o through a structured sequence of calculations.

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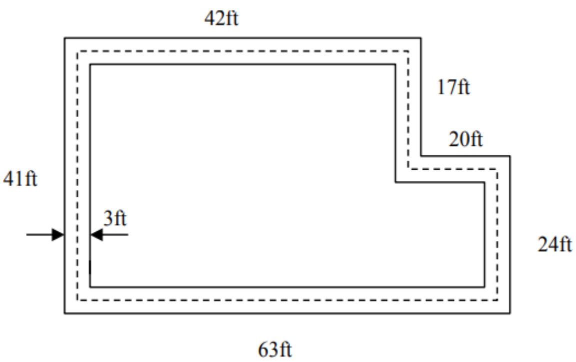
Question #1:



A contractor is excavating the above trench. He is supposed to dig the trench 5 ft deep x 3 ft wide. The soil was tested to have an approximate swell factor of 15% and a shrinkage factor of 12%. The contractor is placing a 8" water pipe in the trench and then backfilling with the soil that was removed. The above dimensions are on centerline.

Does the contractor have enough soil to backfill the trench, or will he/she have to need more? If he needs more soil, how much does he/she need to bring in. Answer in LCY.

Figure 7: Example of excavation calculations in QTO

As shown in Figure 7, this is the task where we chose to validate our proposed method, whose exact answer is shown in Figure 8.

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Step 1: Find the Length of the Trench

- **Trench Length:** `42ft + 17ft + 20ft + 24ft + 63ft + 41ft = 207 ft`

Step 2: Find the Volume of the Soil in the Trench

- **Volume Calculation:** `5ft x 3ft x 207ft = 3105 ft³`

Step 3: Find the Volume after Compaction

- **Compaction Calculation:** `(3105 ft³)(1-0.12) = (3105 ft³)(0.88) = 2732.4 ft³`

Step 4: Find the Volume of the Soil Needed

- **Soil Volume Needed:**
  - `Volume of Trench – Volume of Pipe`
  - `π * (d² / 4) * (207ft) = π * (8/12)² / 4 * (207ft)`
  - `3105 ft³ - 72.25 ft³ = 3032.75 ft³`

Step 5: Find Out if You Need More Soil

- **Additional Soil Required:** `3032.75 ft³ - 2732.4 ft³ = 300.35 ft³`
- **Result:** So the KTR needs to bring in more soil.

Step 6: How Much Soil Does the Contractor Need to Bring In

- **Soil to Bring In:**
  - `Volume of compacted soil = (1 - Shrinkage factor) x BCF`
  - `300.35 ft³ / 0.88 = 341 BCF, LCF = BCF x 1.15 = 392/27 ≈ 14.5 LCY`

Figure 8: Real answer to the question covered in case study

First, we give this question to GPT-4o to complete without any processing. The process and results are as shown in the Figure 9. Since the inference text of GPT-4o is too long when it comes to complex calculations, to facilitate typesetting, we will re-display the exact same content in markdown format.

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Let's solve the problem step by step.

Step 1: Determine the Length of the Trench

- Trench dimensions are given as 5 ft deep and 3 ft wide. The total length is calculated based on the centerline measurements:
  - **Length of the trench:** `63ft + 42ft + 41ft + 17ft + 20ft + 24ft - (2 x 3 ft) = 204 ft`
- **Volume of the trench:**
  - **Length x Depth x Width** = `204 ft x 5 ft x 3 ft = 3060 cubic feet`

Step 2: Calculate the Swelled Volume of Soil

- Considering the swell factor due to soil removal, the volume increases:
  - **Swell factor:** `15%`
  - **Swelled volume:** `3060 cubic feet x 1.15 = 3519 cubic feet`

Step 3: Calculate the Volume Occupied by the Pipe

- The pipe inside the trench has a diameter of 8 inches:
  - **Diameter conversion to feet:** `8 inches / 12 = 0.6667 ft`
  - **Radius of the pipe:** `0.6667 ft / 2 = 0.3333 ft`
  - **Volume per foot of pipe:** `π x (0.3333 ft)² x 1 ft = 0.349 cubic feet per foot`
  - **Total volume of the pipe:** `0.349 cubic feet/foot x 204 ft = 71.1 cubic feet`

Step 4: Calculate the Volume of Soil Needed for Backfilling (Considering Shrinkage Factor)

- The volume of soil required for backfilling is calculated as the volume of the trench minus the volume of the pipe, adjusted for soil shrinkage:
  - **Volume for backfilling:** `(3060 cubic feet - 71.1 cubic feet) x 1 / (1 - 0.12) = 2988.9 cubic feet`

Step 5: Compare the Swelled Soil Volume with the Required Backfill Volume

- Compare the previously calculated swelled soil volume to the backfill volume needed:
  - **Swelled volume:** `3519 cubic feet`
  - **Required backfill volume:** `3396.5 cubic feet`

Since the swelled volume (`3519 cubic feet`) is greater than the required backfill volume (`3396.5 cubic feet`), the contractor has enough soil to backfill the trench. No additional soil is needed.

Conclusion

- The contractor has enough soil to backfill the trench. No additional soil is needed.

Figure 9: The excavation task is answered using GPT-4o without our method

By comparing the GPT-4o answers with the real answer, we can see that the process and results produced by GPT-4o highlighted significant errors in calculations and understanding. GPT-4o failed to parse the complex semantic information and calculations involved in the task, resulting in incorrect results. This demonstrates the limitations of using a general language model for such domain-specific tasks without appropriate preprocessing and specialized methodologies.

To solve the problem in this task, we begin with an exterior construction

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specification related to excavation, but in this specification the obvious formula does not exist, as shown in Figure 10.

Original construction specification on excavation volume (no obvious formula)

According to the Code for Building Earthworks, the calculation of excavation quantities must follow a comprehensive process. First, determine the total length of the excavation by summing all segment lengths and the centerline length. The excavation volume is then calculated by multiplying the specified depth, width, and overall length from the earthworks specifications. This calculated volume should be adjusted by applying a shrinkage factor to account for changes in soil volume after excavation. Additionally, the volume of any installed pipes must be deducted from the total excavated volume according to pipe installation specifications. Once the net excavation volume is determined, it should be compared to the adjusted earthwork volume to identify any deficiencies. If the net excavation volume is less than the adjusted earthwork volume, no additional soil is required. If the net excavation volume exceeds the adjusted earthwork volume, the difference represents the additional earthwork volume needed, which should be divided by the reduction factor. Finally, convert this additional earthwork volume to loose cubic yards by taking into account the expansion coefficient, ensuring the accuracy of material requirements.

Figure 10: External construction specification on excavation volume

Next, we employ GPT-4o to implement the parsing of external construction specifications and to construct the corresponding formula calculation chains. The implementation of this method is illustrated in Figure 11.

Let's solve the problem step by step.

1. Total Length Calculation:
   - According to the construction and earthwork engineering norms, the total length of the excavation should be determined by summing the lengths of all segments and centerline lengths.
   - **Formula:**
     - **Total Length = ∑ Segment Lengths**

2. Volume of Trench Calculation:
   - The volume of the soil to be excavated is calculated by multiplying the depth, width, and total length, as specified in the earthwork engineering norms.
   - **Formula:**
     - **Volume of Trench = Depth × Width × Total Length**

3. Compacted Volume Calculation:
   - The resulting volume should then be adjusted for soil compaction and shrinkage factors, in line with foundation treatment norms, to account for changes in soil volume after excavation.
   - **Formula:**
     - **Compacted Volume = Volume of Trench × (1 − Shrinkage Factor)**

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4. Volume of Pipe Calculation:

- The volume of any installed pipes should be subtracted from the total excavated volume according to pipeline installation norms.
- **Formula:**
  - **Volume of Pipe = π × (Pipe Diameter / 2)^2 × Pipe Length**

5. Adjusted Trench Volume Calculation:

- Subtract the volume of the pipe from the volume of the trench.
- **Formula:**
  - **Adjusted Trench Volume = Volume of Trench − Volume of Pipe**

6. Additional Soil Needed Calculation:

- The additional soil needed is calculated and converted to loose cubic yards (LCY) to ensure accurate material requirements.
- **Formula:**
  - **Additional Soil Needed = (Adjusted Trench Volume − Compacted Volume) / (1 − Shrinkage Factor)**
  - **Bulk Cubic Feet (BCF) of Soil to Bring In = Additional Soil Needed × (1 + Swell Factor)**
  - **Loose Cubic Yards (LCY) = Bulk Cubic Feet (BCF) / 27**

Figure 11: GPT-4o parses external specifications and builds calculation chains

Once the specifications are parsed, we proceed to extract construction data relevant to the project from various documents. This involves identifying and extracting the necessary data, which is then processed using OCR technologies to convert it into a usable digital format. The extracted data includes numerical data, text data, and geometric data, all of which are essential for the QTO calculations. The results of layout analysis are shown in the Figure 12.

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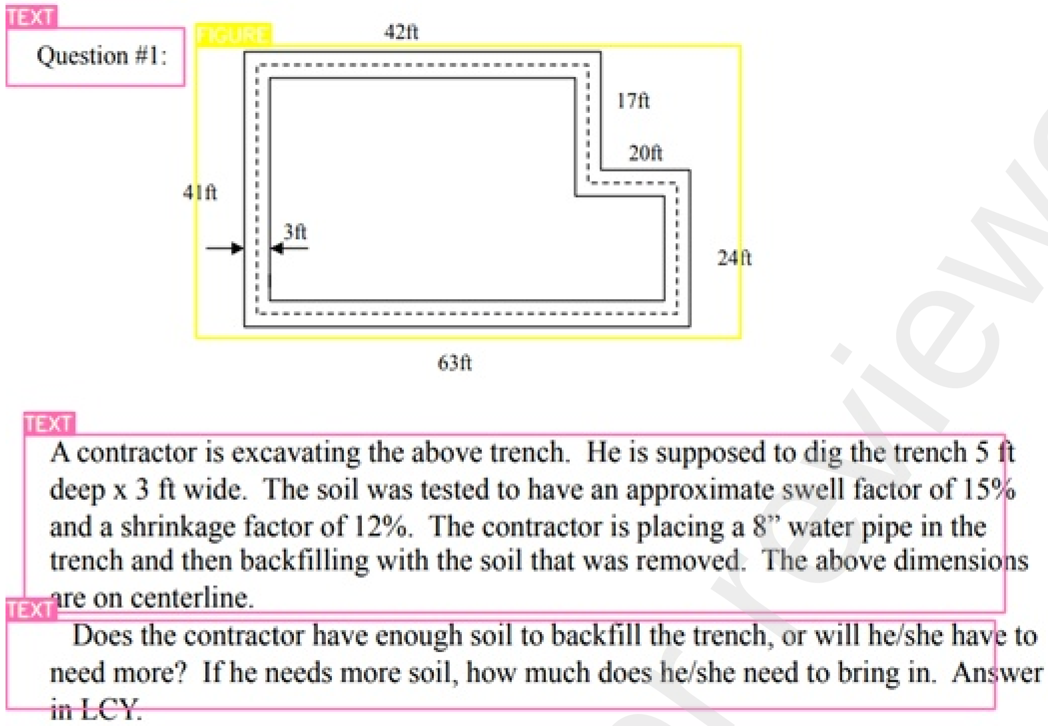

Figure 12: Layout analysis results

The corresponding recognition results are shown in the Figure 13, including text information and numerical information required for QTO calculation. However, some small errors also occurred during the recognition process. For instance, the text "8-inch" was recognized with a slight error, resulting in "8ⁿ " instead of "8" with the inch symbol ("). The geometric information is saved in the form of images. This step ensures that all extracted data is accurately captured and ready for subsequent symbol mapping phase.

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Question #1: 42ft
17ft
20t
41ft
3 24ft
63ft
A contractor is excavating the above trench. He is supposed to dig the trench 5 ft deep x 3 ft wide. The soil was tested to have an approximate swell factor of 15% and a shrinkage factor of 12%. The contractor is placing a 8" water pipe in the trench and then backfilling with the soil that was removed. The above dimensions are on centerline.
Does the contractor have enough soil to backfill the trench, or will he/she have to need more?If he needs more soil, how much does he/she need to bring in.
Answer in LCY.

Figure 13: Text and numerical information extracted from construction data

Next, we input this extracted information, along with the previously established calculation chains and external specification details, into the LLMs for symbol mapping. By leveraging the contextual clues in the text and structured information from external rule documents, LLMs ensure precise and accurate mapping of all extracted data to their respective symbols.

The results of this comprehensive mapping process are shown in Figure 14. Here, we see how the numerical values, such as the lengths of trench segments and soil properties, are mapped to their corresponding symbols in the QTO calculation chains. This includes converting units where necessary and ensuring that each data is correctly linked to its respective symbol. This result demonstrates the effectiveness of our method in associating numerical information with the appropriate symbols for QTO calculations, ensuring that the data is ready for precise and efficient calculation in the following steps.

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Mapped Symbols:

1. Total Length (L)
   - **Extracted Data:** 42ft, 17ft, 20ft, 24ft, 63ft, 41ft
   - **Mapped Symbol:** `L = 207 ft`
2. Depth (D)
   - **Extracted Data:** 5ft
   - **Mapped Symbol:** `D = 5 ft`
3. Width (W)
   - **Extracted Data:** 3ft
   - **Mapped Symbol:** `W = 3 ft`
4. Shrinkage Factor
   - **Extracted Data:** 12%
   - **Mapped Symbol:** `Shrinkage Factor = 0.12`
5. Swell Factor
   - **Extracted Data:** 15%
   - **Mapped Symbol:** `Swell Factor = 0.15`
6. Pipe Diameter
   - **Extracted Data:** 8 inches
   - **Mapped Symbol:** `Pipe Diameter = 8 inches`
7. Pipe Length
   - **Extracted Data:** 207ft
   - **Mapped Symbol:** `Pipe Length = 207 ft`

Figure 14: The result of symbol mapping process

Having mapped the symbols, the next step involves using CoT prompting to guide the LLMs through the QTO calculations. CoT prompting involves breaking down the problem into a series of logical steps, allowing the LLMs to follow a structured reasoning process. This method enhances the accuracy and reliability of the calculations by ensuring that each step is clearly defined and logically follows from the previous one.

For the excavation task, the CoT prompt template is shown in Figure 15.

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Using the established calculation chain, apply the project-specific data to compute the required quantities for the excavation task.

Project Data:

- **Segment Lengths (SL):** 42ft, 17ft, 20ft, 24ft, 63ft, 41ft
- **Depth (D):** 5 ft
- **Width (W):** 3 ft
- **Shrinkage Factor (SF):** 0.12
- **Swell Factor (SW):** 0.15
- **Pipe Diameter (PD):** 8 inches
- **Pipe Length (PL):** 207 ft

Calculation Chain:

1. Total Length Calculation:

- According to the construction and earthwork engineering norms, the total length of the trench should be determined by summing the lengths of all segments and centerline lengths.
- **Formula:**
  - **Total Length = ∑ Segment Lengths**

2. Volume of Trench Calculation:

- The volume of the soil to be excavated is calculated by multiplying the depth, width, and total length, as specified in the earthwork engineering norms.
- **Formula:**
  - **Volume of Trench = Depth × Width × Total Length**

3. Compacted Volume Calculation:

- The resulting volume should then be adjusted for soil compaction and shrinkage factors, in line with foundation treatment norms, to account for changes in soil volume after excavation.
- **Formula:**
  - **Compacted Volume = Volume of Trench × (1 − Shrinkage Factor)**

4. Volume of Pipe Calculation:

- The volume of any installed pipes should be subtracted from the total excavated volume according to pipeline installation norms.
- **Formula:**
  - **Volume of Pipe = π × (Pipe Diameter / 2)^2 × Pipe Length**

5. Adjusted Trench Volume Calculation:

- Subtract the volume of the pipe from the volume of the trench.
- **Formula:**
  - **Adjusted Trench Volume = Volume of Trench − Volume of Pipe**

6. Additional Soil Needed Calculation:

- The additional soil needed is calculated and converted to loose cubic yards (LCY) to ensure accurate material requirements.
- **Formula:**
  - **Additional Soil Needed = (Adjusted Trench Volume − Compacted Volume)**
  - **Bulk Cubic Feet (BCF) of Soil to Bring In = Additional Soil Needed / (1 − Shrinkage Factor)**
  - **Loose Cubic Yards (LCY) = Bulk Cubic Feet (BCF) × (1 + Swell Factor) / 27**

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Step-by-Step Calculations

1. Total Length Calculation:

- Sum the lengths of the trench segments: 42 ft, 17 ft, 20 ft, 24 ft, 63 ft, and 41 ft.
- **Calculation:**
  - `Total Length = 42 + 17 + 20 + 24 + 63 + 41 = 207 ft`

2. Volume of Trench Calculation:

- Calculate the volume of the trench using the formula: Depth × Width × Total Length.
- The depth is 5 ft and the width is 3 ft.
- **Calculation:**
  - `Volume of Trench = 5 ft × 3 ft × 207 ft = 3105 ft³`

3. Compacted Volume Calculation:

- Adjust the volume for a shrinkage factor of 12%. Use the formula: Volume × (1 - Shrinkage Factor).
- **Calculation:**
  - `Compacted Volume = 3105 ft³ × (1 - 0.12) = 2732.4 ft³`

4. Volume of Pipe Calculation:

- Calculate the volume of the 8-inch diameter pipe using the formula for the volume of a cylinder: π × (Diameter/2)² × Length. Convert the diameter to feet.
- **Calculation:**
  - `Volume of Pipe = π × (8 in / 2 / 12 ft)² × 207 ft = 72.25 ft³`

5. Adjusted Trench Volume Calculation:

- Subtract the volume of the pipe from the volume of the trench.
- **Calculation:**
  - `Adjusted Trench Volume = 3105 ft³ - 72.25 ft³ = 3032.75 ft³`

6. Additional Soil Needed Calculation:

- Determine if additional soil is needed by comparing the adjusted trench volume with the compacted volume.
- **Calculation:**
  - `Additional Soil Needed = 3032.75 ft³ - 2732.4 ft³ = 300.35 ft³`

7. Convert to Loose Cubic Yards (LCY):

- Calculate the additional soil needed in loose cubic yards (LCY) using the swell factor of 15% and converting to LCY.
- **Calculation:**
  - `Bulk Cubic Feet (BCF) of Soil to Bring In = 300.35 ft³ / 0.88 = 341.31 ft³`
  - `Loose Cubic Yards (LCY) = 341.31 BCF × 1.15 / 27 ≈ 14.54 LCY`

Figure 15: CoT prompt template for excavation task

This CoT template provides a structured and detailed approach for guiding LLMs through the QTO calculations for excavation tasks. By comparing the calculation process performed by LLM with the calculation process of the real answer, it demonstrates how LLM flawlessly completed all the calculations, highlighting the superiority of our method, especially in handling complex computations.

By meticulously following the outlined steps—calculating the total length of the

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trench, determining the volume of the trench, adjusting for soil compaction and shrinkage factors, calculating the volume of the pipe, and finally computing the additional soil needed for backfilling—the LLMs have proven their capability in performing detailed and precise calculations. This accuracy not only emphasizes the robustness of our proposed method but also showcases the efficiency and reliability of LLMs in automating complex construction-related tasks.

In stark contrast, the results obtained directly using GPT-4o highlighted significant errors in calculations and understanding. Our proposed method effectively handled the complex QTO calculations by systematically parsing the rules, accurately extracting relevant data, mapping data and corresponding symbols, and employing advanced reasoning methods. This improvement marks a significant advancement in the field of automated construction management, indicating the potential for enhanced efficiency and accuracy in project planning and resource management.

To further verify the effectiveness of the proposed method on different types of QTO tasks, we created an open dataset of thirty questions. This dataset includes three most common types of QTO tasks, namely earthwork, masonry and rebar work. We first tested this dataset using GPT-4o without any pre-processing or specialized methods, and the results showed that GPT-4o failed to complete these QTO tasks correctly with 0% accuracy. This result demonstrates that directly applying general LLMs to domain-specific tasks (such as QTO) can lead to significant errors, especially without the support of additional domain knowledge and structured methods. Next, we applied the proposed method, which combines data extraction, symbol mapping, and rule-based reasoning, and tested it on the same dataset using GPT-4o. With this method, we significantly improved our accuracy, getting 24 out of 30 questions correct, as shown in Table. 1. Our method performed flawlessly on earthwork and masonry work but encountered challenges with rebar work. The issues in the rebar work likely came from the complexity of the images involved, which require advanced geometric understanding and deeper, more specialized domain knowledge. This result

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demonstrates that while our method significantly enhances GPT-4o's capability in QTO tasks, particularly with simple tasks, the limitations in handling more complex tasks like rebar work highlight the need for further improvements. By raising GPT-4o's accuracy to 80%, our method still shows great potential for automating QTO in construction projects using traditional 2D documents.

Table. 1 Performance comparison in QTO tasks using different methods

Methods \ QTO tasks	Earthwork	Masonry work	Rebar work
GPT-4o	0/12	0/12	0/6
GPT-4o with proposed method	12/12	12/12	0/6

5. Discussions and Conclusions

The methodology presented in this study demonstrates significant advancements in automating QTO from traditional construction documents using LLMs. One of the main strengths of this approach is its ability to significantly enhance the speed and accuracy of QTO processes compared to traditional manual methods and even some BIM-based methods. By integrating advanced data extraction techniques, symbol mapping based on LLMs and rule-based structured CoT reasoning, our method ensures that complex QTO processes can be calculate correctly.

However, there are limitations to this approach. The accuracy of the automated QTO is heavily reliant on the quality of the input data, including the external construction specifications and the OCR technologies employed. Any errors in these inputs can propagate through the pipeline, leading to inaccuracies in the final QTO results. Additionally, while the case study validated the method's effectiveness for excavation tasks, its applicability to other types of construction tasks remains to be further explored and validated.

Another potential limitation is the computational resources required for

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implementing LLMs with CoT reasoning. The method may pose challenges for real-time applications in large-scale construction projects due to the processing power needed. Future research could address these issues by exploring the integration of more efficient models or hybrid approaches that combine LLMs with other emerging technologies, such as IoT and computer vision, to further enhance the capabilities and applicability of the proposed method.

In conclusion, this study presents a novel and robust approach to automating QTO processes using LLMs. The main contributions of this research are as follows:

- **Enhanced the ability of LLMs to interpret and reason with human language in construction-specific tasks**: This research significantly extends the application of LLMs by improving their ability to understand and process the complex language used in construction documents. This advancement moves LLMs beyond basic language comprehension to deep, task-specific reasoning, contributing to the broader field of LLMs' application in construction domains.
- **Developed a comprehensive method to automate QTO processes**: This method can automatically extract data from various unstructured files and accurately map symbols into construction formulas. This not only greatly reduces the time and effort of manual QTO, but also makes complex calculation steps more logical and transparent by introducing chain reasoning. Each calculation step is traceable and auditable, ensuring the consistency and accuracy of QTO calculations in demanding construction projects.
- **Expanded the practical applicability of LLM-based automation in non-BIM construction environments**: The proposed method provides an efficient solution for automating QTO in non-BIM environments, expanding the practical applications of LLMs in a wider range of construction projects. This contribution addresses a critical gap in the industry, offering a scalable, adaptable alternative to BIM-dependent processes.

Overall, this research advances the state of the art in automated QTO, providing a

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practical and scalable solution that leverages the power of LLMs to address the challenges of traditional QTO methods.

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